

What can we learn from option prices?

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Microsoft (MSFT) – Underlying Asset

MARKET SNAPSHOT (MAY 2026)

Stock price
~**\$422**
NASDAQ: MSFT

Market cap
\$3.0T
4th largest

FY2025 Revenue
\$282B
+15% YoY

AI ARR
\$37B
+123% YoY

KEY BUSINESS DRIVERS

- **Azure cloud:** +40% YoY in Q3 FY2026, first quarter crossing \$54.5B in Microsoft Cloud revenue
- **Intelligent Cloud segment:** \$34.7B revenue (+30% YoY)
- **OpenAI partnership:** \$100B+ invested, exclusive license through 2032

WHY MSFT FOR DERIVATIVES?

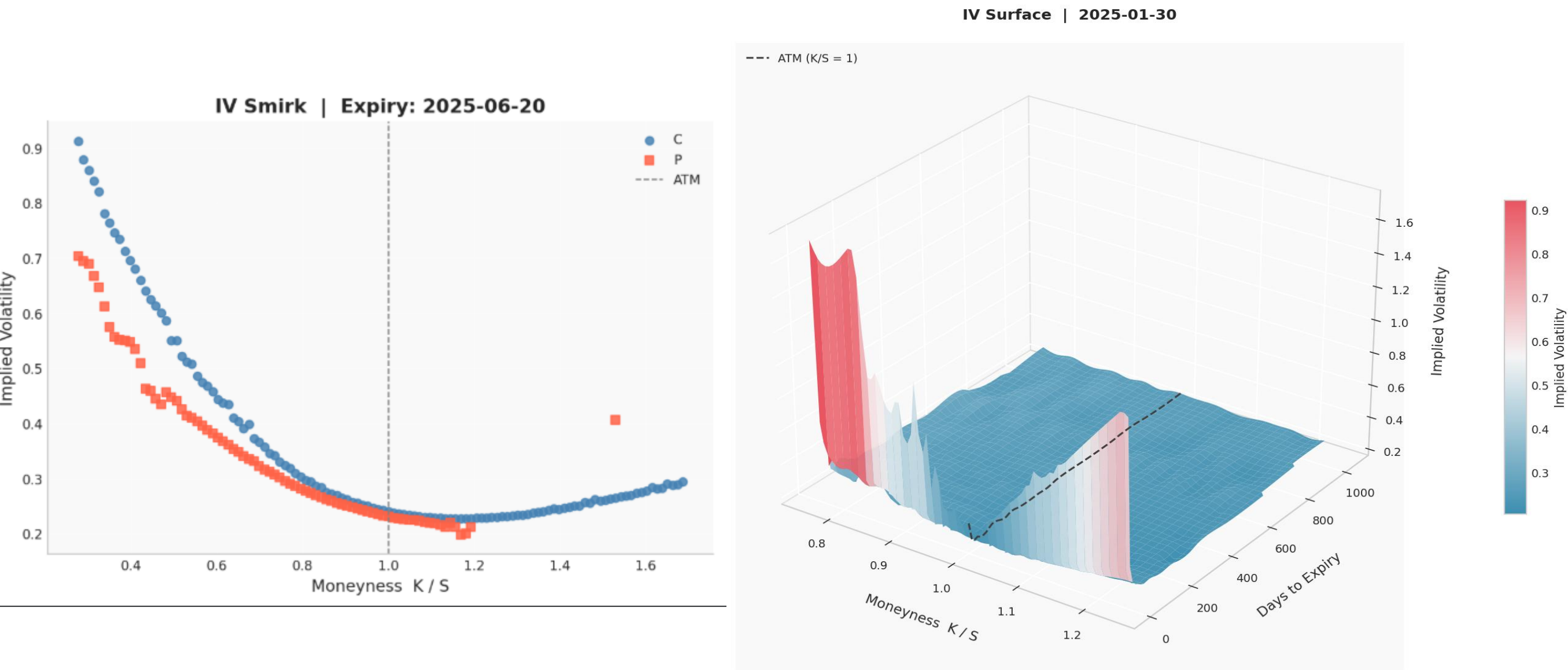
High information flow: Earnings, Azure guidance and AI news generate sharp, well-identified uncertainty shocks, directly observable in implied volatility around event dates.

Exceptional options liquidity: MSFT options trade across many strikes and maturities, including weekly expirations. This lets us observe rich volatility smiles, estimate risk-neutral densities and run delta-hedging in a realistic setting.

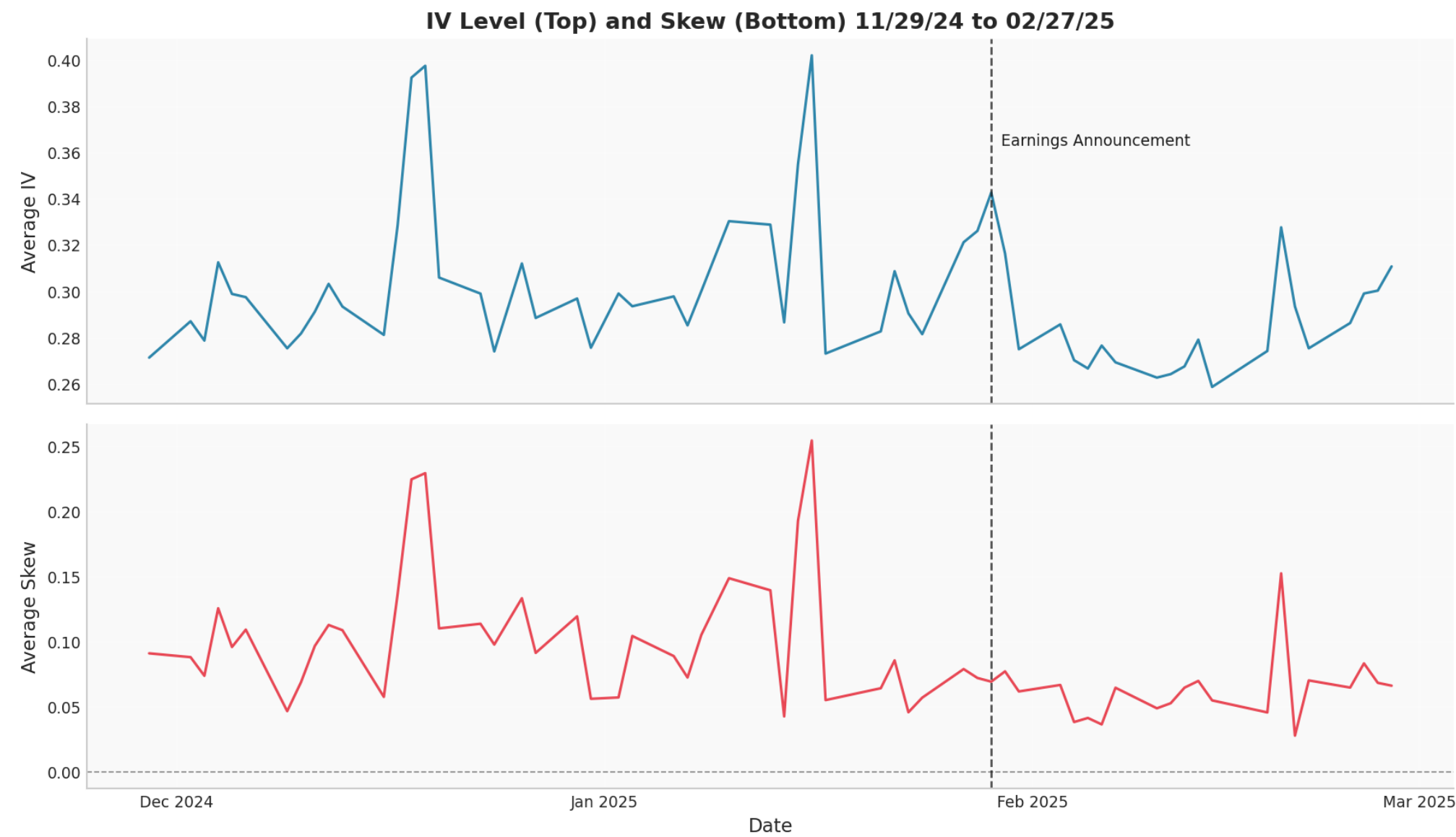
Strong institutional coverage: 38 analysts, "Strong Buy" consensus. Deep two-sided market ensures tight bid-ask spreads and reliable implied volatility data.

Implied Volatility

- The 30th of January 2025 is used to construct our implied volatility surface.
- The uneven volatility surface means the Black-Scholes assumption of a constant volatility (σ) in both days to expiry and moneyness is violated in markets.
- Plotting the IV-Moneyness relationship with puts and calls separate helps visualise the volatility smirk - OTM puts trade at a higher IV than ITM calls.
- OTM puts are typically used for hedging downside risk, so the smirk can be seen as the premium that investors are willing to pay in to hedge against left-tail risk.



- Across the dataset, average IV and Skew tend to move together in response to events.
- They both spike together in Dec, mid-Jan and Feb – unforeseen waves of uncertainty. But around the earnings announcement only IV jumps.
- This de-linking captures the symmetric uncertainty of earnings announcements – investors brace for both left and right tail risk.



Risk-Neutral Density

Dates (same expiry):

1) Jan 30, 2025 - post-earnings, peak volume ($T \approx 0.88$ y, $S = 415$)

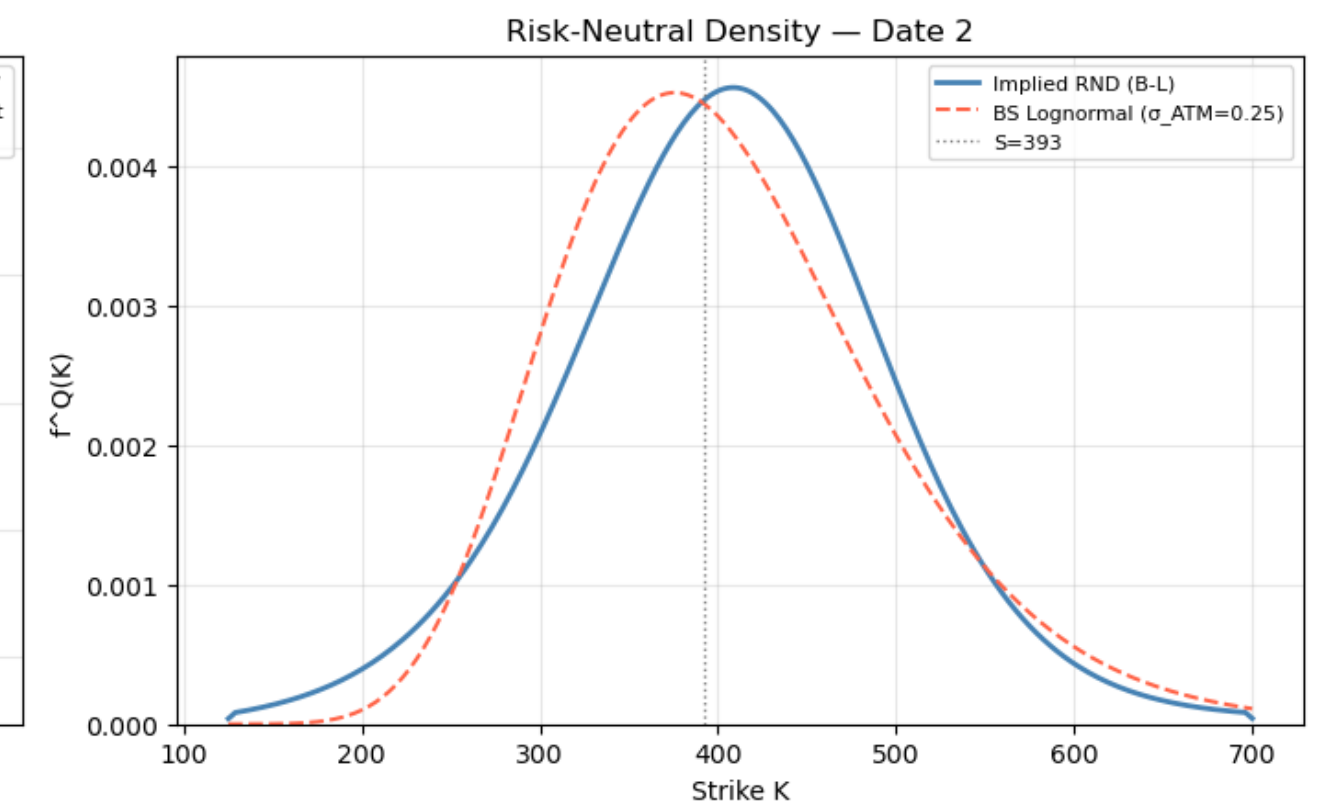
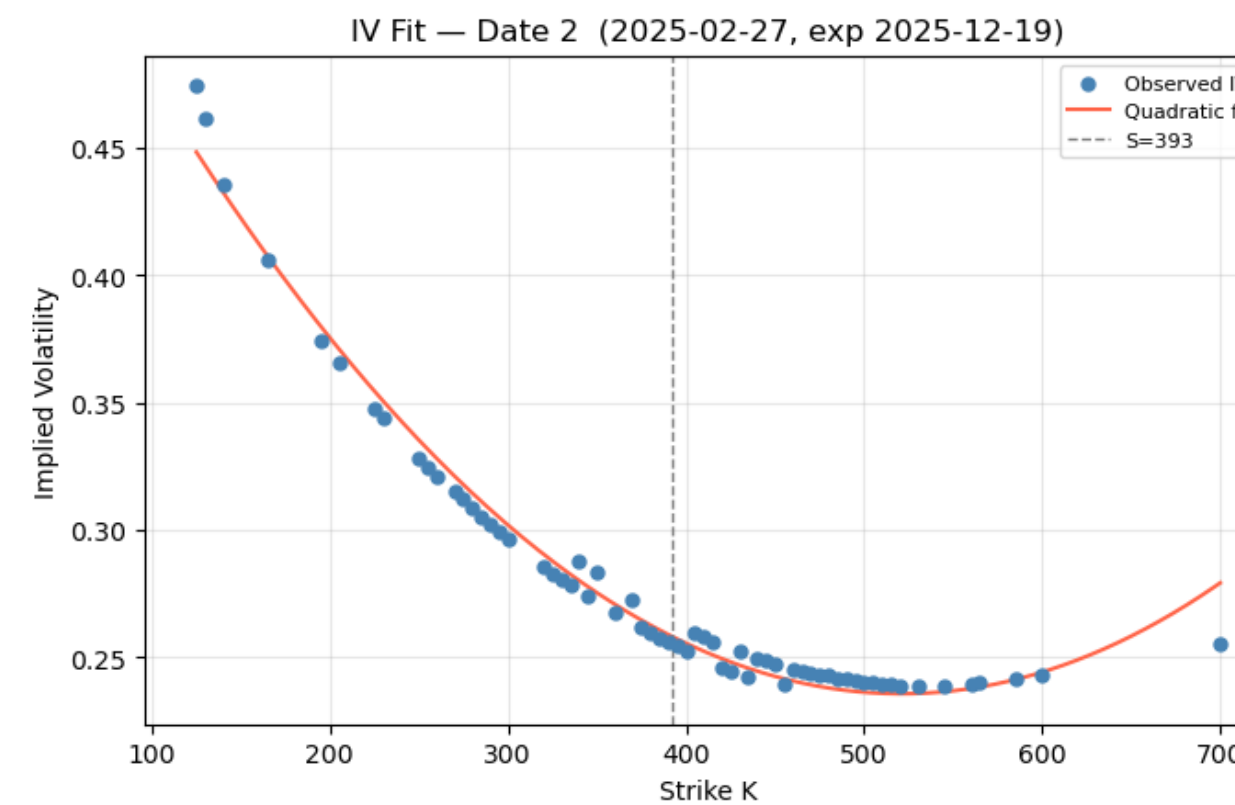
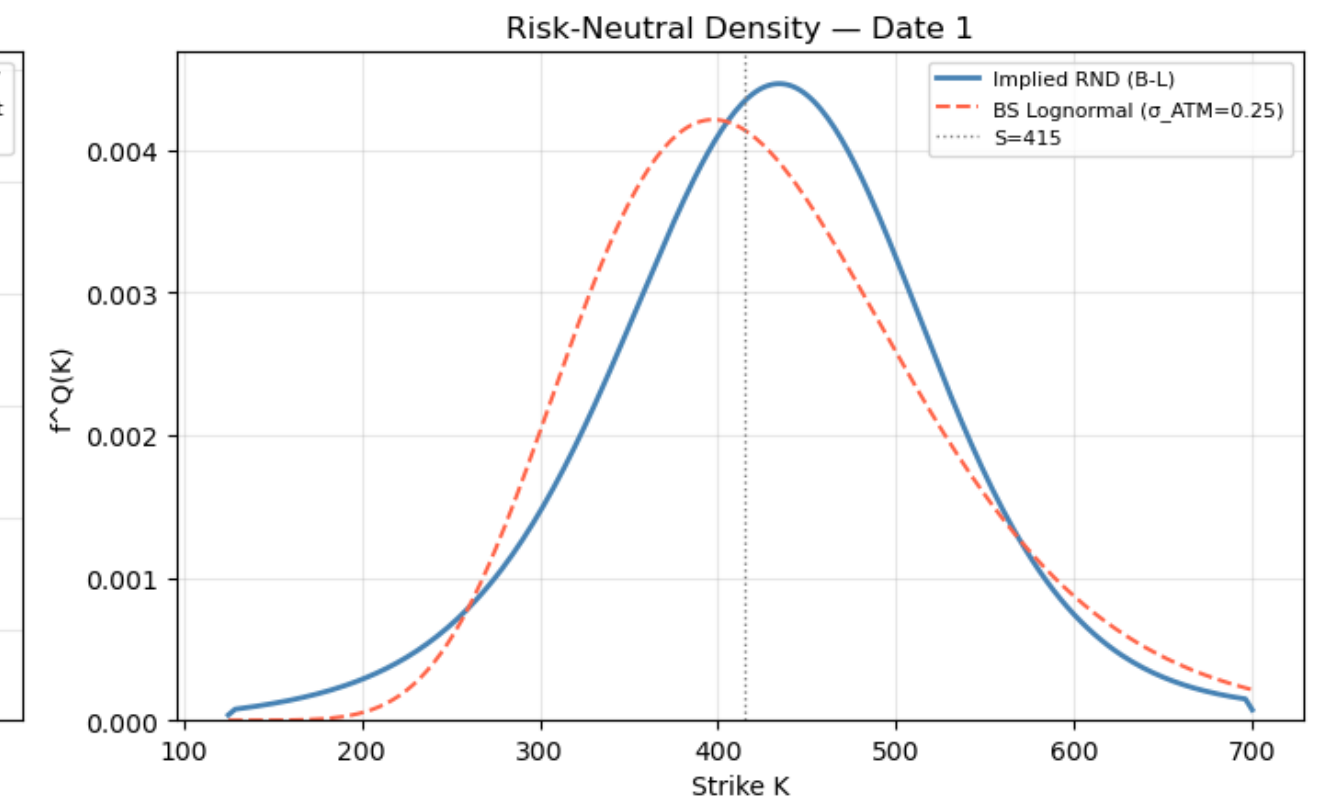
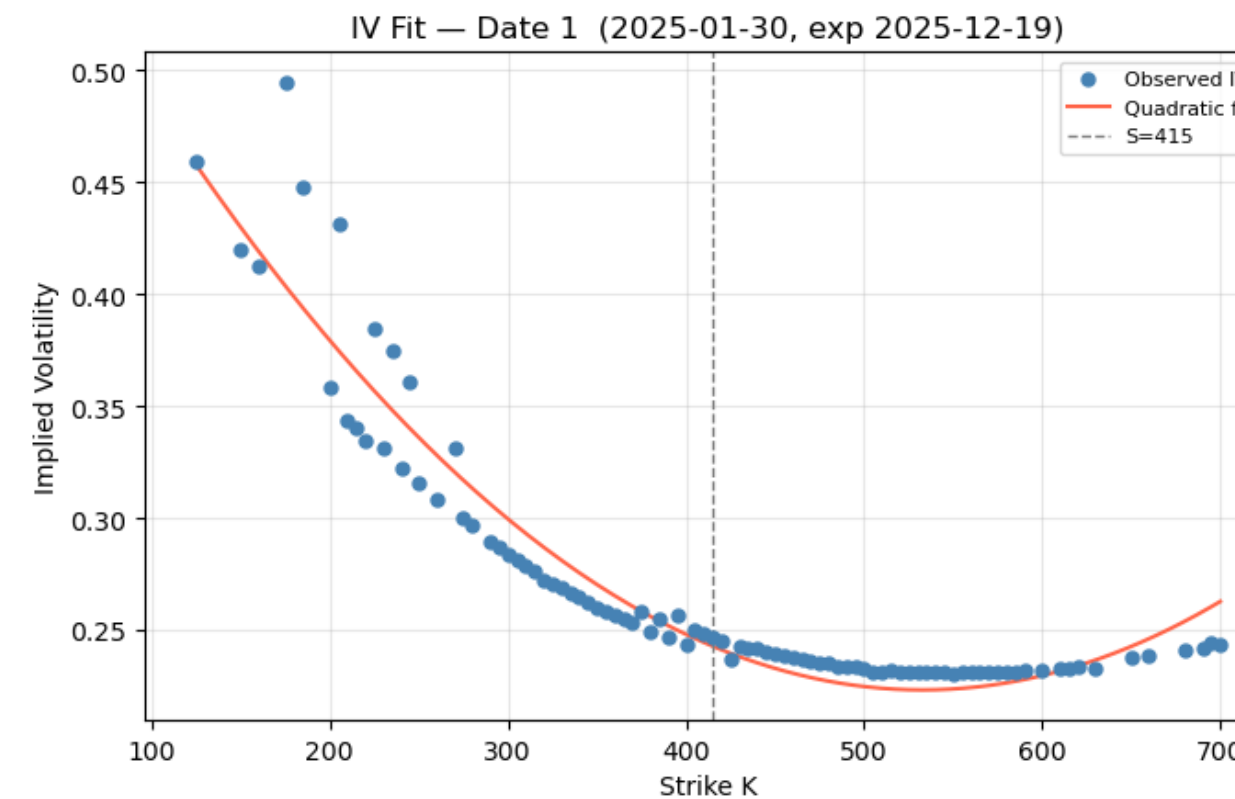
2) Feb 27, 2025 - last day in dataset ($T \approx 0.81$ y, $S = 393$)

Methodology:

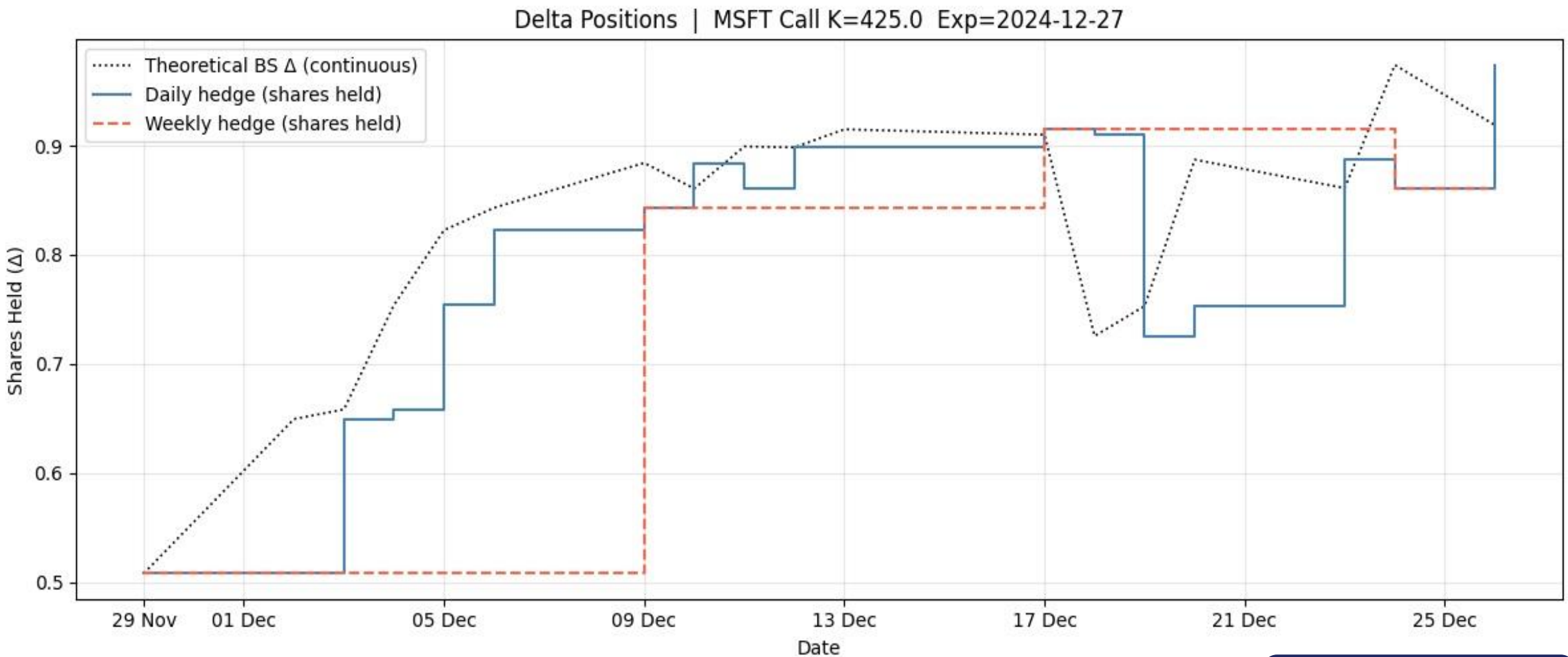
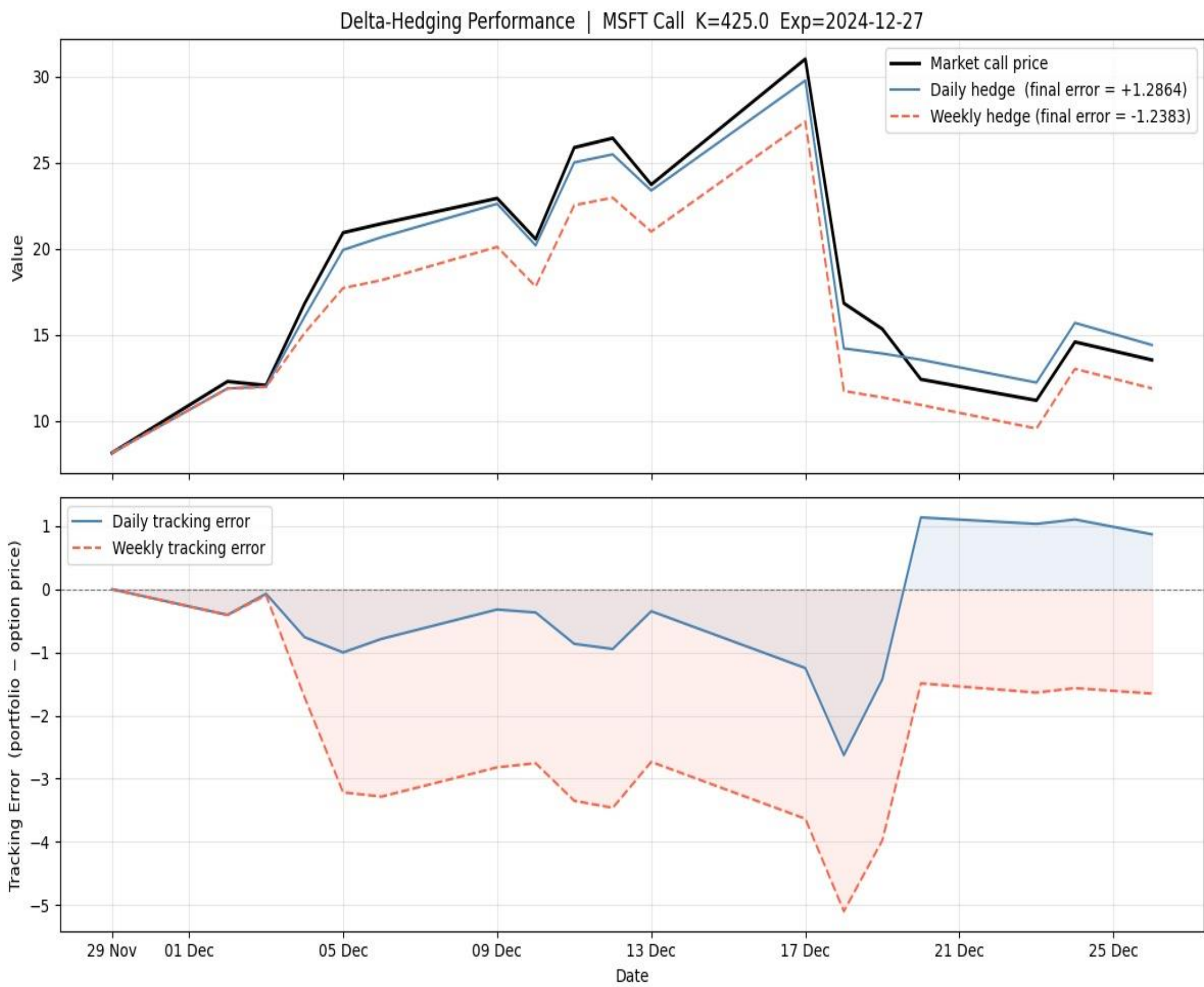
- Fitting a quadratic polynomial to IV as a function of moneyness to reduce impact of price errors on $\text{std}(\widehat{C''(K)})$
- Moneyness (K/S) instead of strike (K) because of conditioning problems
- Computing $C(K)$ using Black-Scholes and fitted IV
- Applying Breeden-Litzenberger ($f^Q(K) = e^{rT} \frac{\partial C(K)}{\partial K^2}$) to approximate the second derivative and building the Risk-Neutral Density

Results:

- Implied Risk-Neutral Density is less right-skewed than Black-Scholes log-normal, meaning market prices excess downside risk.
- In the second date the smile flatter, and the implied density is closer to log-normal, meaning that the earnings shock has been digested.



Delta-Hedging



Option selected: Near-ATM MSFT call (K=425, ~28 days).
ATM maximises gamma, hardest case to hedge.

Daily error +10.2%: Portfolio over-performs, trending rally causes slight delta over-accumulation.

Weekly error -9.1%: Portfolio under-performs; lagged rebalancing misses the growing delta.

Main error sources: Discrete gamma P&L, vol mismatch (entry IV vs. realised), and Black-Scholes model risk.

Daily Error
+1.29
+10.2% of payoff

Weekly Error
-1.24
-9.1% of payoff

Payoff
13.11
vs. premium 8.125



Conclusion

*Option prices are a lens from which we can see the **market's** perspective of the future, but uncertainty is irreducible.*

01

Options look forward

Unlike stock prices (backward-looking), option prices encode the market's risk-neutral distribution over future states of the world.

02

Models are languages, not truths

Black-Scholes is empirically rejected every-day (by volatility smiles), yet whole industry quotes in BS units; it provides a common scale on which heterogeneous option prices can be quoted, compared, and traded..

03

No Arbitrage outlives the model

Breeden-Litzenberger recovers the risk-neutral density from market prices alone, showing that no-arbitrage survives the failure of the parametric assumptions on which Black-Scholes itself depends.

04

The gap is information, not error

The smile, the skew, the hedging residuals are **not** BS "failing"; they are the market revealing structure (jumps, stoch. vol, risk premia) that the model abstracts away.

Limitations

1. Trading costs & Bid-ask Spreads (among others)

A daily hedge over one month means ~20 round trips, and those costs compound. The "optimal" hedging frequency in practice is interior, not maximal - daily isn't strictly better than weekly once you net it out.

2. Single-instrument hedging

We hedged the option using just the underlying stock. Real desks Vega-hedge using other options and Gamma-hedge with shorter-dated contracts.

3. The aggregate ≠ the future (a CLT caveat)

Even if the market's view aggregates thousands of participants, which by CLT logic is the best collective forecast available, it remains a forecast, not a reality. Uncertainty is irreducible; no amount of aggregation beats it.